

RAJ DESAI

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SUMMARY

Software Engineer with 2 years of hands-on experience shipping distributed systems and AI-powered tools, from an 8-service distributed platform to automated code review bots. Specializing in React, Python, TypeScript, and async architectures. MS in Computer Science from UT Dallas, with a build-first, figure-it-out approach to every problem.

EDUCATION

The University of Texas at Dallas | MS – Computer Science Dallas, TX | May 2026
Relevant Courses: Design Analysis of Algorithm, Machine Learning, NLP, AI, Big Data Analytics, Database Design.
University of Mumbai | BE – Computer Engineering Mumbai, India | May 2024

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, Dart, C/C++, SQL, HTML, CSS
Frontend: React, Next.js, Tailwind CSS, Flutter, Framer Motion, Recharts
Backend: Node.js, Express.js, FastAPI, Celery, REST APIs, Redis, BullMQ, RabbitMQ, OAuth 2.0, JWT
Databases: PostgreSQL, pgvector, MySQL, MongoDB, Prisma ORM, SQLAlchemy
ML & AI: LLMs (OpenAI, Gemini APIs), LiteLLM, FAISS, Sentence Transformers, Pandas, NumPy
Tools & Platforms: Git, GitHub Actions, Docker, Docker Compose, CI/CD, OpenTelemetry, Vitest, Vercel, Railway

PROFESSIONAL EXPERIENCE

Software Engineer Intern - Hydracoral Technologies Jan 2023 – Dec 2023 | Mumbai, India

- Accelerated user verification by 30% by developing a Flutter/Dart onboarding flow with facial-recognition APIs, validated in staging
- Reduced app crash rate by 20% by diagnosing and resolving 15+ bugs in state management, API error handling, and memory leaks across the Flutter codebase using Firebase Crashlytics and structured debugging workflows
- Improved usability scores by 20% by designing responsive cross-platform UI components validated through structured user-testing
- Owned CI/CD pipelines in Git and Bitbucket for a 3-person Agile team, integrating REST APIs for JWT authentication, data sync, and release automation across mobile and backend services

PROJECTS

ClauseGuard - AI-Powered Contract Review Platform Apr 2026 – May 2026

- Designed and deployed an 8-service distributed system (FastAPI, Celery, PostgreSQL/pgvector, Redis, RabbitMQ, MinIO) orchestrated via Docker Compose with health checks, graceful degradation, and OpenTelemetry distributed tracing
- Engineered a saga-pattern pipeline across Celery and RabbitMQ with circuit breakers on LLM calls, fan-out parallelism for clause classification, and Redis Pub/Sub SSE for live frontend progress
- Reduced LLM costs by 35–50% via semantic caching with pgvector cosine similarity, structured JSON outputs via Pydantic, and model-tier routing through a pluggable LiteLLM fallback layer.

Yoru - Cinematic Ambient Music Streaming Web App Jan 2026 – Feb 2026

- Shipped a cinematic streaming app with Next.js 15, TypeScript (strict), and Tailwind CSS, orchestrating 21 scenes with Framer Motion crossfades and dual-channel Howler.js audio mixing
- Engineered custom React hooks (useKeyboard, useIdleDetection, useAudio), encapsulating reusable interaction logic with WCAG 2.1 AA keyboard navigation, ARIA support and screen-reader compatibility
- Maintained >80% test coverage using Vitest and React Testing Library, enforcing component-driven architecture with strict TypeScript, zero any types, and isolated unit tests for each custom hook

PR Sensei - AI-Powered GitHub Code Review App Nov 2025 – Jan 2026

- Automated pull-request reviews to under 30 seconds by developing an AI GitHub App that posts summaries and inline comments via OpenAI/Gemini APIs, with a pluggable LLM provider layer triggered by webhook events
- Solved GitHub's 10s webhook timeout against 5–15s LLM calls by engineering an async pipeline with Redis and BullMQ, handling retries, rate limiting, and graceful degradation
- Centralized review analytics and file-hotspot tracking across repos by designing a multi-tenant PostgreSQL schema with Prisma ORM, served to a React dashboard with Recharts visualizations and per-repository filtering

LLM Cookbook - AI-Based Semantic Recipe Search Mar 2025 – Apr 2025

- Delivered sub-1-second semantic search over 10,000+ recipes using FAISS-indexed MiniLM embeddings served through a FastAPI backend, validated across 200 test queries
- Enabled natural-language recipe discovery by designing RESTful API endpoints with multi-dimensional filtering by diet, allergens, and cuisine, powered by Sentence Transformers with ranked relevance scoring